

What the records look like

Five templates for the records we hand over. They show the structure and the acceptance questions; they contain no project data. Project records are shared with the owner, and with others only with the owner's permission.

1. Factory acceptance test record

One sheet per prefabricated assembly (skid, rack manifold or CDU), completed at the factory before release. Gate 1 of commissioning.

Project	Assembly and serial	Drawing and revision	Date
Tested by	Witnessed by		

TEST	REFERENCE	ACCEPTANCE CRITERION	RESULT	PASS / FAIL
Hydrostatic pressure test	Procedure ref.	Test pressure ___ kPa held ___ min; no visible leak; pressure decay within ___ %		
Leak test of joints and QDs	Procedure ref.	No weep at test pressure; method stated (bubble, tracer gas or decay)		
Cleanliness of wetted surfaces	Sample ref.	Particle count at or below the agreed class, verified by sample		
Torque of threaded and flanged joints	Torque table ref.	All joints to table; wrench calibration certificate attached		
Quick-disconnect register	Register ref.	Every QD listed by serial with mate cycles made during factory testing		

TEST	REFERENCE	ACCEPTANCE CRITERION	RESULT	PASS / FAIL
Controls function check	Checklist ref.	Pump staging, valve travel and sensor readings as specified		
Document check	Index ref.	Drawings at stated revision, material certificates, test sheets signed		

2. Turnover package index

The first page of the package. Three questions for each of the ten leak points: was it tested, against what, and where is the record.

#	BOUNDARY	WAS IT TESTED?	AGAINST WHAT CRITERION?	WHERE IS THE RECORD?	STATUS
01	Cold plate joints At the server				Open / closed
02	Quick disconnects At the server				Open / closed
03	Hose crimps At the server				Open / closed
04	Manifold branches In the rack				Open / closed
05	Elastomeric seals In the rack				Open / closed
06	Threaded, compression and adapter fittings In the rack				Open / closed

#	BOUNDARY	WAS IT TESTED?	AGAINST WHAT CRITERION?	WHERE IS THE RECORD?	STATUS
07	Internal boundaries In the CDU				Open / closed
08	Field welds and hot work Between the CDU and the facility				Open / closed
09	Balancing and isolation valves Between the CDU and the facility				Open / closed
10	Fill, drain, vent and sample ports Between the CDU and the facility				Open / closed

3. System boundary diagram

Drawn for each project, so every party can see where its responsibility starts and stops. The six boundaries follow the thermal path.

1 Cold plate and chip Cold-plate maker, server OEM Record owner: _____	2 Hoses, crimps and fittings Hose assembler, rack integrator Record owner: _____	3 Rack manifold and quick disconnects Rack integrator, QD supplier Record owner: _____	4 CDU and secondary loop CDU maker, integrator Record owner: _____	5 Facility water interface Facility mechanical contractor Record owner: _____	6 Heat rejection and reuse Heat-rejection vendor or heat buyer Record owner: _____
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4. Flow-balance test record, with its conditions

A reading without its conditions cannot be compared with anything. The conditions are recorded with the results. Gate 3 of commissioning.

IT load at test (%)		Facility supply T (°C)		Ambient dry bulb (°C)		Date and time		
Instruments and calibration		Witnessed by						
RACK	DESIGN FLOW (L/MIN)	MEASURED FLOW (L/MIN)	DEVIATION (%)	TOLERANCE (%)	SUPPLY T (°C)	RETURN T (°C)	ΔP (KPA)	PASS / FAIL
R01								
R02								
R03								

5. Non-conformance and re-test log

Every finding is raised, fixed, re-tested and closed in writing. Open items at handover are listed, owned and dated, and accepted by the owner. Gate 4 of commissioning.

ID	BOUNDARY	FINDING	RAISED BY, DATE	CORRECTIVE ACTION	OWNER	RE-TEST RESULT	CLOSED BY, DATE
NC-001							
NC-002							
NC-003							